2 Generate Fake data

2. Solve for the equilibrium prices for each good j in each market t.

Consider a snap shot of the generated market data. please see the full DGP in the attached python file

Table 1: Market Data Sample

Index	Price	Share	Market ID	Firm ID	Satellite	Wired	x	w
0	2.764 784	0.086 328	0	0	1	0	0.253 759	0.717 270
1	2.643 251	0.200 549	0	1	1	0	0.134 024	0.000 864
2	2.827 237	0.227 401	0	2	0	1	0.140 025	1.290 341
3	2.603 836	0.133 281	0	3	0	1	0.107 216	0.368 745
4	3.008 030	0.397 030	1	0	1	0	1.560 624	1.403 483
				⋮				
2395	2.707 730	0.013 148	598	3	0	1	0.414 766	0.915 487
2396	2.396 050	0.261 938	599	0	1	0	1.218 003	0.246 415
2397	3.720 164	0.011 428	599	1	1	0	0.954 761	2.082 937
2398	2.736 636	0.026 670	599	2	0	1	0.186 165	0.931 112
2399	3.372 414	0.553 740	599	3	0	1	2.295 622	0.049 518

Note: Table shows first and last five observations from 2400 total entries. Satellite and Wired are binary indicators (1=Yes, 0=No).

3. Calculate "observed" shares for your fake data set using your parameters, your draws of x,w, β, ω , ξ, and your equilibrium prices.

Consider a snap shot of the generated market data. please see the full DGP in the attached python file

Table 2: Market-Level Data Sample (Abridged)

Market ID	Firm ID	Omitted Variables[1]	ξ	ω	Price ($)	δ	exp (δ)	Obs Share
0	0	⋮	−0.294 793	0.106 181	2.764 784	−1.570 603	0.207 920	0.000
0	1	⋮	0.424 771	0.102 034	2.643 251	−0.727 706	0.483 016	0.000
0	2	⋮	0.973 698	−0.707 799	2.287 237	−0.540 751	0.382 311	0.000
0	3	⋮	0.025 453	0.098 975	2.603 835	−1.075 670	0.341 297	0.000
1	0	⋮	0.821 574	−0.152 805	3.008 030	0.366 138	1.442 154	0.000
				⋮				
598	3	⋮	−1.926 513	0.058 749	2.707 730	−2.927 206	0.053 546	0.000
599	0	⋮	0.008 362	−0.123 380	2.396 050	0.434 165	1.543 671	0.000
599	1	⋮	−0.212 349	0.145 881	3.720 164	−2.097 917	0.067 346	0.000
599	2	⋮	−0.320 911	0.065 602	2.736 636	−1.608 017	0.209 824	0.000
599	3	⋮	1.874 361	0.172 978	3.372 414	1.425 156	4.158 505	0.000

[1] Omitted variables (x, w, Sat., Wired) are reported in Table 2. ξ = demand shock, ω = cost shock

[2] Obs Share = Observed Market Share (trimmed to 0.000 for display)

3 Estimate Some Mis-specified Models

1. Estimate the plain multinomial logit model of demand by OLS (ignoring the endogeneity of prices).

Table 3: OLS Regression Results

Dep. Variable:	log_odds	**R-squared:**	0.275
Model:	OLS	**Adj. R-squared:**	0.274
Method:	Least Squares	**F-statistic:**	302.6
Date:	Thu, 27 Mar 2025	**Prob (F-statistic):**	1.50e-166
Time:	23:01:43	**Log-Likelihood:**	-3178.1
No. Observations:	2400	**AIC:**	6364.
Df Residuals:	2396	**BIC:**	6387.
Df Model:	3	**Covariance Type:**	nonrobust

	coef	**std err**	**t**	**P** $> \|t\|$	**[0.025**	**0.975]**
x	0.8395	0.032	26.181	0.000	0.777	0.902
satellite	1.2104	0.133	9.087	0.000	0.949	1.472
wired	1.2039	0.134	8.951	0.000	0.940	1.468
prices	-0.9381	0.049	-19.318	0.000	-1.033	-0.843

Omnibus:	16.355	**Durbin-Watson:**	1.926
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	16.482
Skew:	-0.199	**Prob(JB):**	0.000264
Kurtosis:	3.080	**Cond. No.**	31.0

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

2. Re-estimate the multinomial logit model of demand by two-stage least squares, instrumenting for prices with the exogenous demand shifters x and excluded cost shifters w. Discuss how the results differ from those obtained by OLS.).

Table 4: IV-2SLS Estimation Summary

Dep. Variable:	log_odds	**R-squared:**	0.1447
Estimator:	IV-2SLS	**Adj. R-squared:**	0.1436
No. Observations:	2400	**F-statistic:**	2319.6
Date:	Thu, 27 Mar 2025	**P-value (F-stat):**	0.0000
Time:	23:02:25	**Distribution:**	chi2(4)
Cov. Estimator:	robust		

	Parameter	**Std. Err.**	**T-stat**	**P-value**	**Lower CI**	**Upper CI**
x	0.9579	0.0348	27.541	0.0000	0.8898	1.0261
satellite	3.8668	0.1971	19.615	0.0000	3.4805	4.2532
wired	3.8872	0.1990	19.533	0.0000	3.4971	4.2772
prices	-1.9447	0.0736	-26.436	0.0000	-2.0889	-1.8005

Endogenous: prices
Instruments: w
Robust Covariance (Heteroskedastic)
Debiased: False

The 2SLS addressed the endogeneity associated with price. The estimator has a value of ($\hat{\alpha} = -1.9447$), which is very close to the true parameter value ($\alpha = -2$). However, due to the model misspecification, the estimates on the rest of the attributes remains biased. This is due to the fact that the plain logit model fails to factor other unobserved heterogeneities at individual levels.

4 Estimate the Correctly Speced Model

1. Use the pyBLP package to estimate the correctly speciffied model. Report a table with the estimates of the demand parameters and standard errors. Do this three times:

(a) once when you estimate demand alone,

Table 5: Demand Mixed Logit Estimation Summary

Nonlinear Coefficient Estimates (Robust SEs in Parentheses)		
	satellite	wired
Sigma:		
satellite	1.349 252	
	$(4.026\,017 \times 10^{-1})$	
wired	0.000 000	1.638 966
		(1.111 603)

Beta Estimates (Robust SEs in Parentheses)			
x	prices	satellite	wired
1.035 641	−2.078 766	4.141 683	4.088 462
$(1.381\,446 \times 10^{-1})$	$(2.714\,851 \times 10^{-1})$	$(6.358\,548 \times 10^{-1})$	$(4.474\,204 \times 10^{-1})$

- **There are 3 instruments in total**
  - A **Demand** instrument that using supply shifter W for the price endogeneity
  - BLP **Demand** instrument that is constructed based on sum of other products' characteristics in the market, assuming single-product firms
  - BLP **Demand** differentiated instrument is constructed on the pairwsie product difference across all markets, assuming for single-product firms

(b) Then again when you estimate demand jointly with supply

Table 6: Demand & Supply Mixed Logit Summary

Nonlinear Coefficient Estimates (Robust SEs in Parentheses)		
	satellite	wired
Sigma:		
satellite	$5.863\,815 \times 10^{-1}$	
	$(2.630\,459)$	
wired	$0.000\,000$	$1.105\,002$
		$(2.526\,125)$

Beta Estimates (Robust SEs in Parentheses)			
x	prices	satellite	wired
$9.454\,731 \times 10^{-1}$	$-1.915\,824$	$3.859\,118$	$3.783\,456$
$(1.047\,116 \times 10^{-1})$	$(2.427\,087 \times 10^{-1})$	$(8.462\,556 \times 10^{-1})$	$(3.038\,551 \times 10^{-1})$

Gamma Estimates (Robust SEs in Parentheses)	
1	w
$4.953\,485 \times 10^{-1}$	$2.471\,269 \times 10^{-1}$
$(2.355\,622 \times 10^{-2})$	$(9.453\,930 \times 10^{-3})$

- **There are 5 instruments in total**
  - A **Demand** instrument that using supply shifter W for the price endogeneity
  - BLP **Demand** instrument that is constructed based on sum of other products' characteristics in the market, assuming single-product firms
  - BLP **Demand** differentiated instrument is constructed on the pairwsie product difference across all markets, assuming for single-product firms
  - BLP **Supply** differentiated instrument is constructed on the pairwsie product difference across all markets, assuming for single-product firms
  - BLP **Supply** differentiated instrument is constructed on the pairwsie product difference across all markets, assuming for single-product firms

(c). again when you estimate demand alone but now with the "optimal IV"

Table 7: Demand Mixed Logit (Optimal IVs)

Nonlinear Coefficient Estimates (Robust SEs in Parentheses)		
	satellite	wired
Sigma:		
satellite	1.291 755	
	$(2.517\,807 \times 10^{-1})$	
wired	0.000 000	1.421 176
		$(2.287\,317 \times 10^{-1})$

Beta Estimates (Robust SEs in Parentheses)			
x	prices	satellite	wired
1.009 364	−2.035 339	4.041 539	4.030 194
$(4.966\,596 \times 10^{-2})$	$(1.033\,566 \times 10^{-1})$	$(2.513\,953 \times 10^{-1})$	$(2.484\,137 \times 10^{-1})$

Table 8: Comparison of Mixed Logit Models with True Parameter Values

Parameter	**True Value**	**Estimates (Robust SEs)**		
		Demand (3 IVs)	**Demand & Supply (5 IVs)**	**Optimal IVs**
Beta Parameters				
x (β_1)	1	1.0356 (0.1381)	0.9455 (0.1047)	1.0094 (0.0497)
prices (α)	-2	−2.0788 (0.2715)	−1.9158 (0.2427)	−2.0353 (0.1034)
satellite (β_2)	4	4.1417 (0.6359)	3.8591 (0.8463)	4.0415 (0.2514)
wired (β_3)	4	4.0885 (0.4474)	3.7835 (0.3039)	4.0302 (0.2484)
Sigma Parameters				
satellite	1	1.3493 (0.4026)	0.5864 (2.6305)	1.2918 (0.2518)
wired	1	1.6390 (1.1116)	1.1050 (2.5261)	1.4212 (0.2287)
Gamma Parameters				
γ_1	0.5	-	0.4953 (0.0236)	-
γ_2	0.25	-	0.2471 (0.0095)	-

Notes: Standard errors in parentheses. Gamma parameters only estimated in joint demand-supply model. True values: $\beta_2, \beta_3 \sim N(4,1)$, $\sigma_{satellite} = \sigma_{wired} = 1$.

Based on the summary table, the optimal IVs model has estimate values (column 5) that are the closest to the true parameter values (column 2)

2. Using your preferred estimates from the prior step (explain your preference), provide a table comparing the estimated own-price elasticities (averaged across markets)to the true own-price elasticities.

Table 9: Comparison of Estimated and True Elasticities by Firm

Firm	**Estimated Elasticity**	**True Elasticity**
Firm 1	−4.076 697	−4.476 395
Firm 2	−4.078 727	−4.460 344
Firm 3	−4.098 088	−4.518 768
Firm 4	−4.096 522	−4.527 721

The estimated elasticities are all around -4.07 to -4.10, while the true elasticities are slightly more negative, ranging from -4.46 to -4.53. The optimal IV's framework suggests a less elastic supply than the true value. The differences aren't huge, but they're noticeable.